Discrete Mathematics and Graph Theory

Graphs

Department of Electronics and Telecommunication Engineering

Ref: Kenneth Rosen, Discrete Mathematics and its Applications, 7th Edition, McGraw Hill Publications, 2012

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Graphs

Module 5

- Graph types (Finite and Infinite graphs, Multigraphs, Directed and Undirected graphs), Graph models
- Graph terminology, Connectivity, Euler and Hamilton paths, Bipartite Graphs, Planar Graphs
- Representing graphs, Adjacency matrices, Incidence matrices, Graph coloring
- Graph Isomorphism

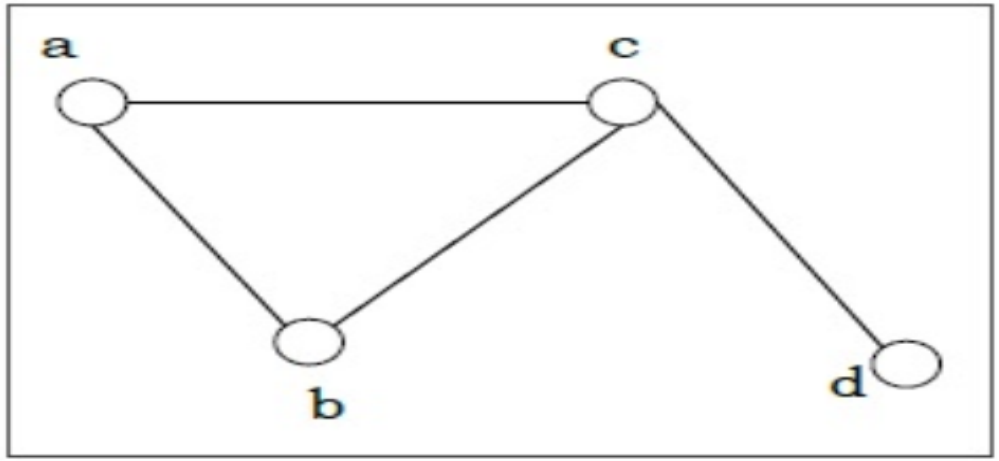
Graphs

A graph is a set of points, called nodes or vertices, which are interconnected by a set of lines called edges. The study of graphs, or graph theory is an important part of a number of disciplines in the fields of mathematics, engineering and computer science.

What is a Graph?

Definition – A graph (denoted as $G = (V, E)$) consists of a non-empty set of vertices or nodes V and a set of edges E.

Example – Let us consider, a Graph is $G = (V, E)$ where $V = \{a, b, c, d\}$ and $E = \{\{a, b\}, \{a, c\}, \{b, c\}, \{c, d\}\}$



Graphs contd…

Degree of a Vertex – The degree of a vertex V of a graph G (denoted by deg (V)) is the number of edges incident with the vertex V.

Vertex	Degree	Even / Odd
a	2	even
b	2	even
c	3	odd
d	1	odd

Graphs contd…

Even and Odd Vertex – If the degree of a vertex is even, the vertex is called an even vertex and if the degree of a vertex is odd, the vertex is called an odd vertex.

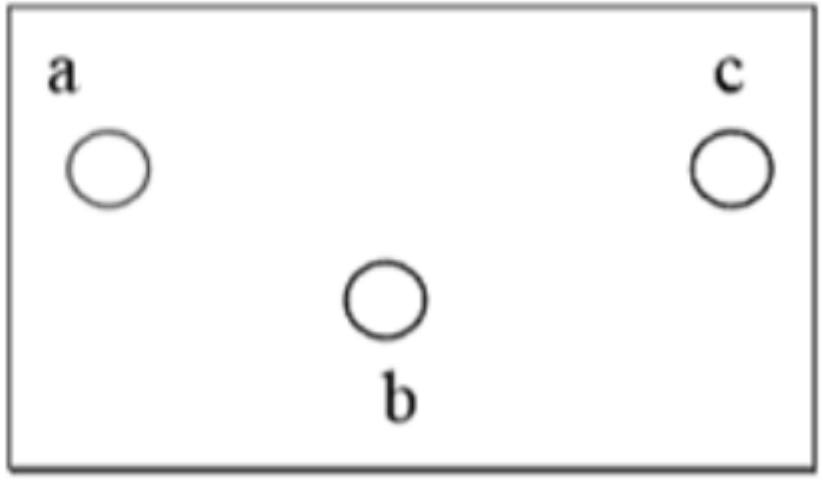
Degree of a Graph – The degree of a graph is the largest vertex degree of that graph. For the previous graph the degree of the graph is 3.

The Handshaking Lemma – In a graph, the sum of all the degrees of all the vertices is equal to twice the number of edges.

Types of Graphs

Null Graph

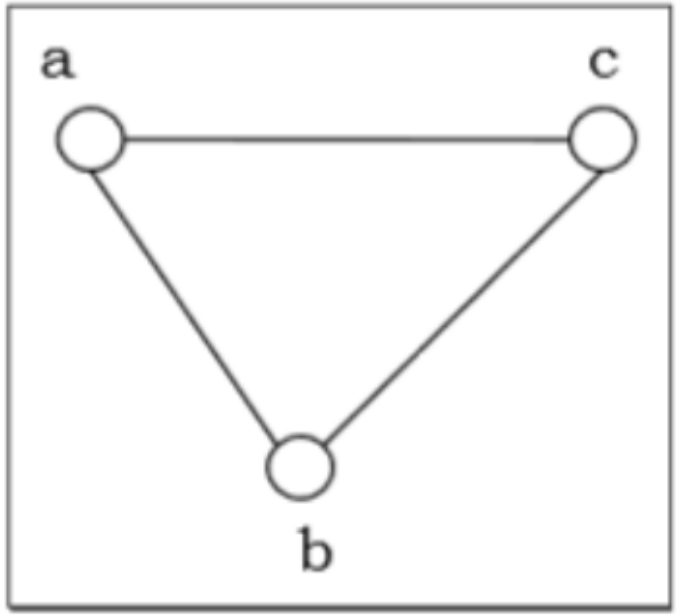
A null graph has no edges. The null graph of n vertices is denoted by N_n



Types of Graphs

Simple Graph

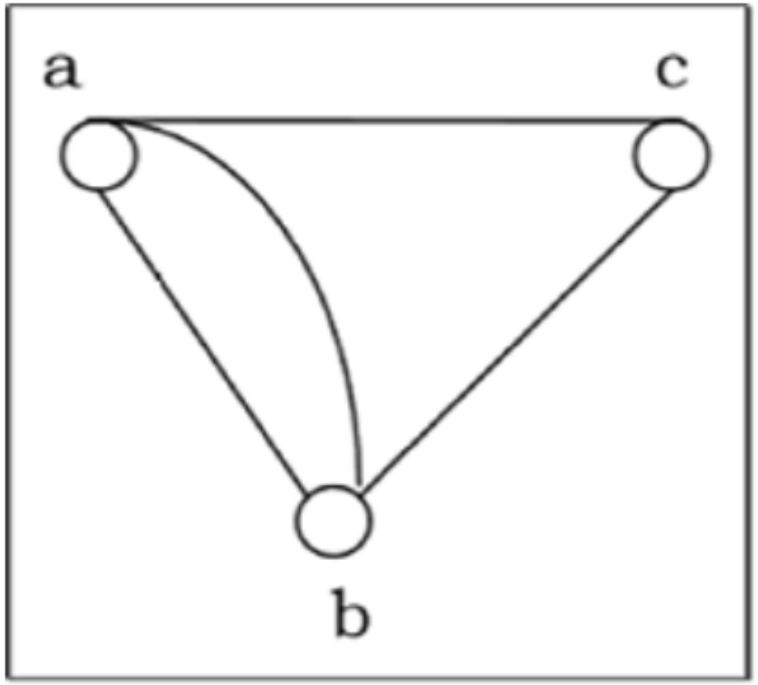
A graph is called simple graph/strict graph if the graph is undirected and does not contain any loops or multiple edges.



Types of Graphs

Multi-Graph

If in a graph multiple edges between the same set of vertices are allowed, it is called Multigraph. In other words, it is a graph having at least one loop or multiple edges.

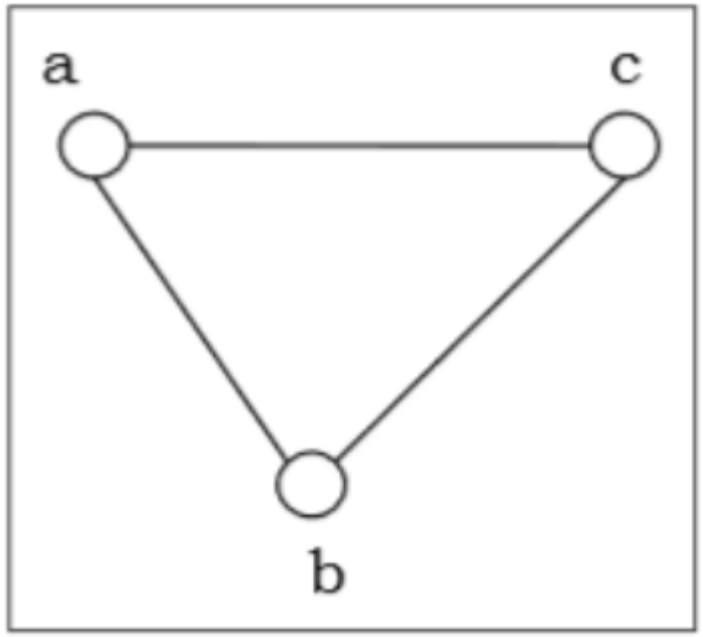


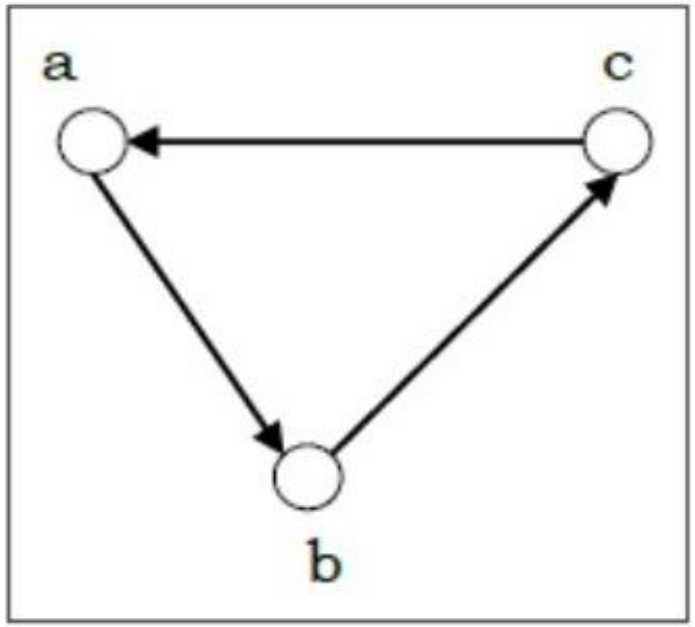
Types of Graphs

Directed and Undirected Graph

A graph $G = (V, E)$ is called a directed graph if the edge set is made of ordered vertex pair and a graph is called undirected if the edge set is made of unordered vertex pair.

Undirected Graph





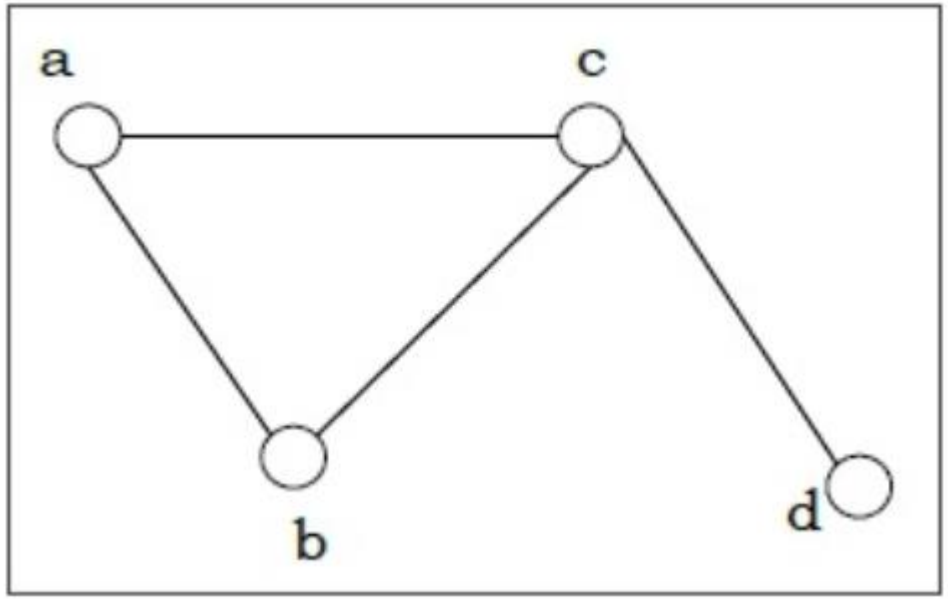
Directed Graph

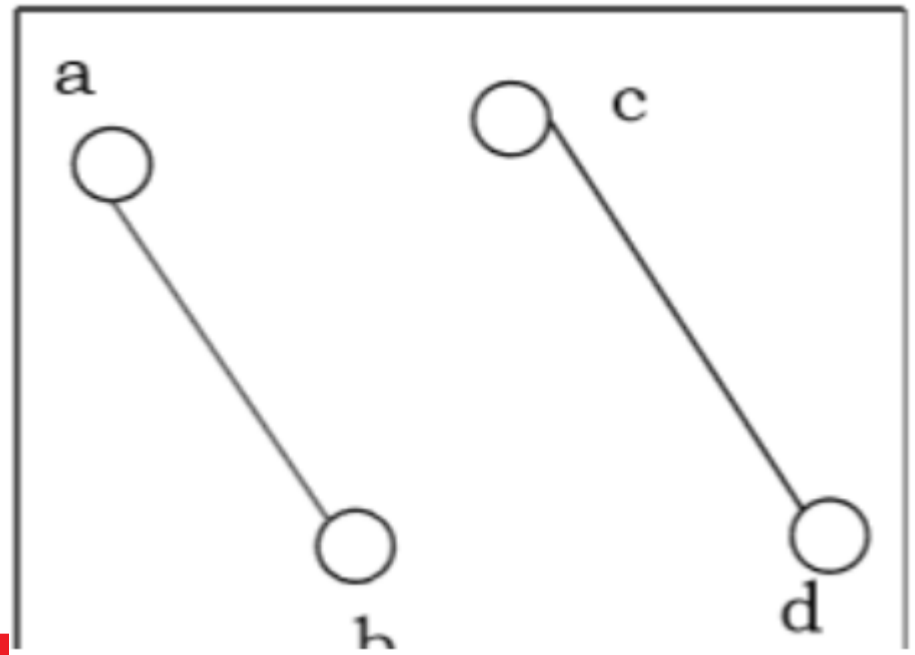
Types of Graphs

Connected and Disconnected Graph

A graph is connected if any two vertices of the graph are connected by a path; while a graph is disconnected if at least two vertices of the graph are not connected by a path. If a graph G is disconnected, then every maximal connected subgraph of G is called a connected component of the graph G.

Connected Graph





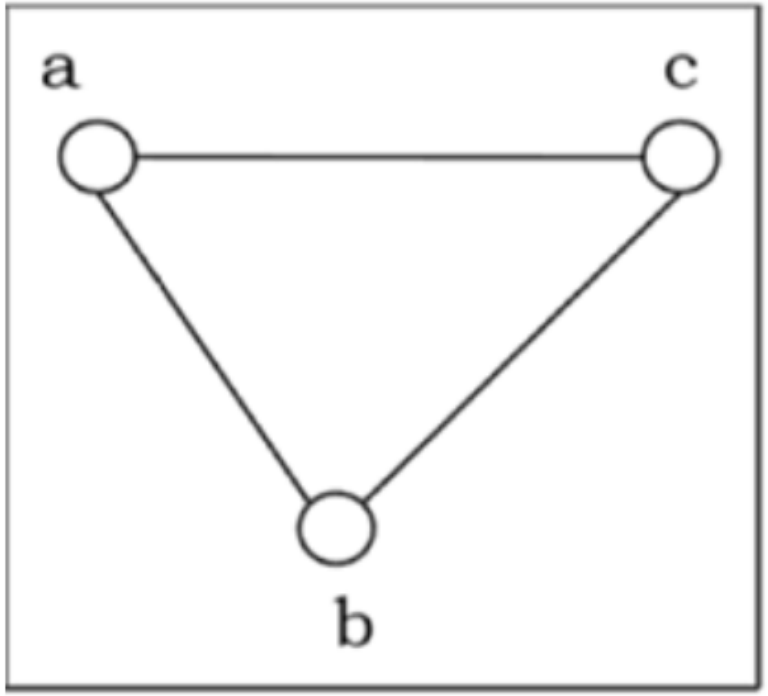
Disconnected Graph

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Types of Graphs

Complete Graph

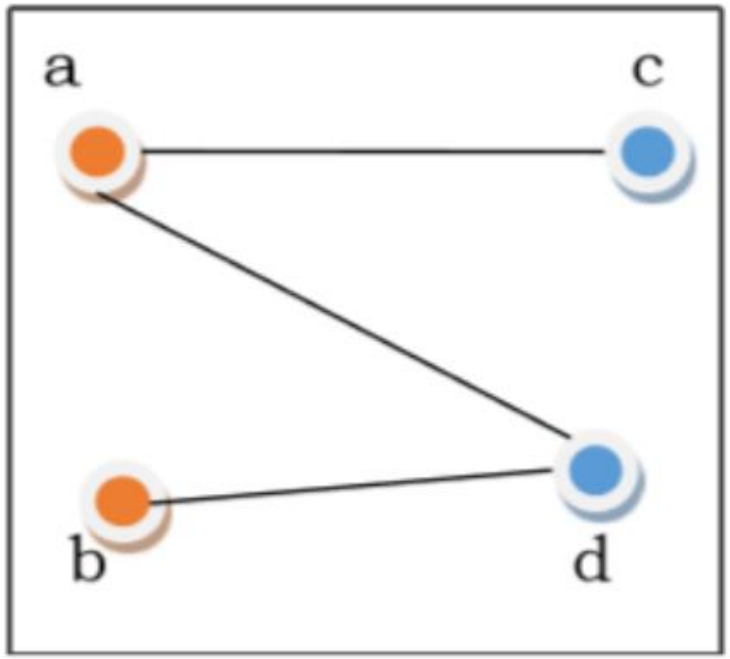
A graph is called complete graph if every two vertices pair are joined by exactly one edge. The complete graph with n vertices is denoted by K_n



Types of Graphs

Bipartite Graph

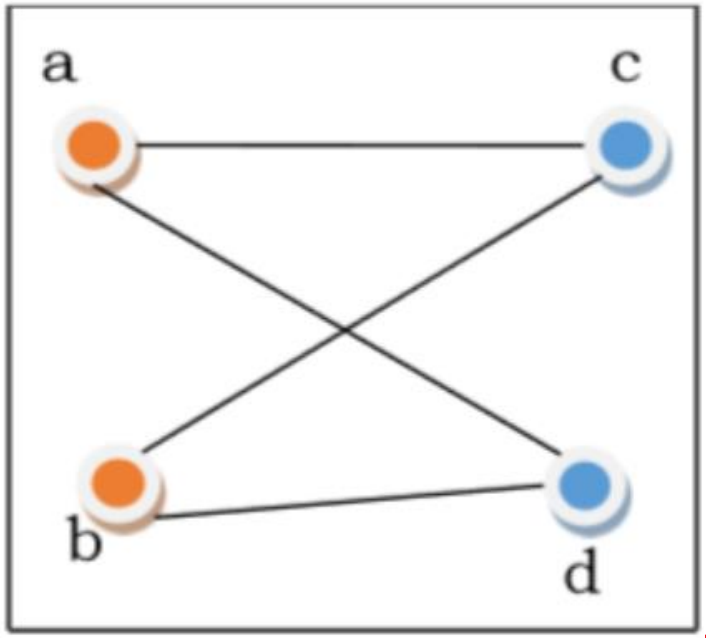
If the vertex-set of a graph G can be split into two disjoint sets, V_1 and V_2, in such a way that each edge in the graph joins a vertex in V_1 to a vertex in V_2, and there are no edges in G that connect two vertices in V_1 or two vertices in V_2, then the graph G is called a bipartite graph.



Types of Graphs

Complete Bipartite Graph

A complete bipartite graph is a bipartite graph in which each vertex in the first set is joined to every single vertex in the second set. The complete bipartite graph is denoted by $K_{x,y}$ where the graph G contains x vertices in the first set and y vertices in the second set.



Graph Terminology

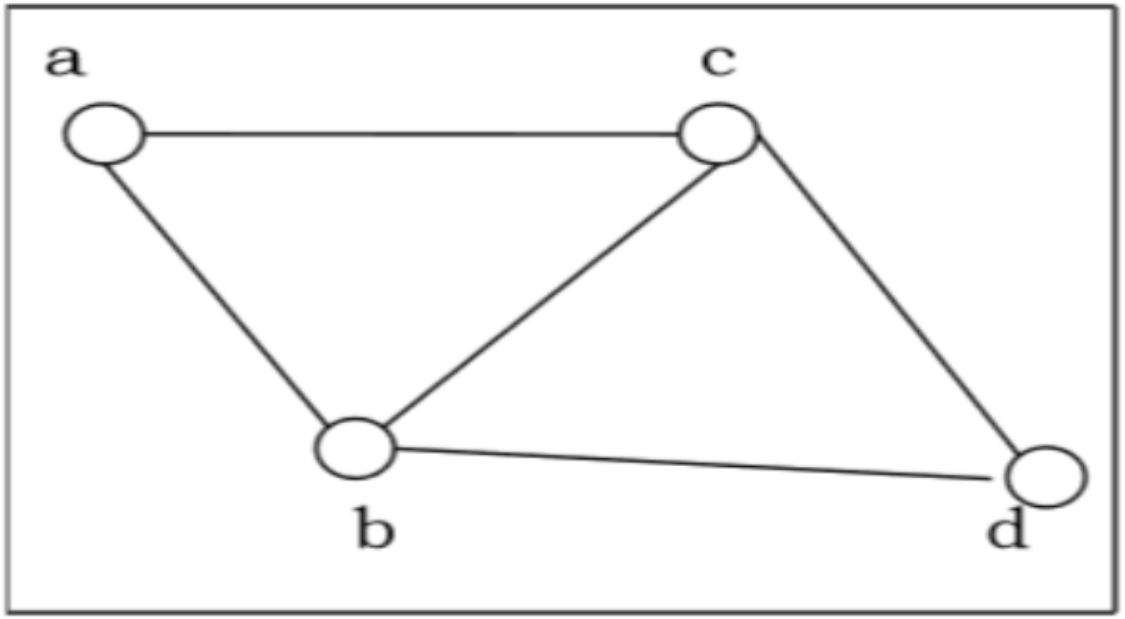
TABLE 1 Graph Terminology.

Type	*Edges*	*Multiple Edges Allowed?*	*Loops Allowed?*
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

Types of Graphs

Planar vs. Non-planar graph

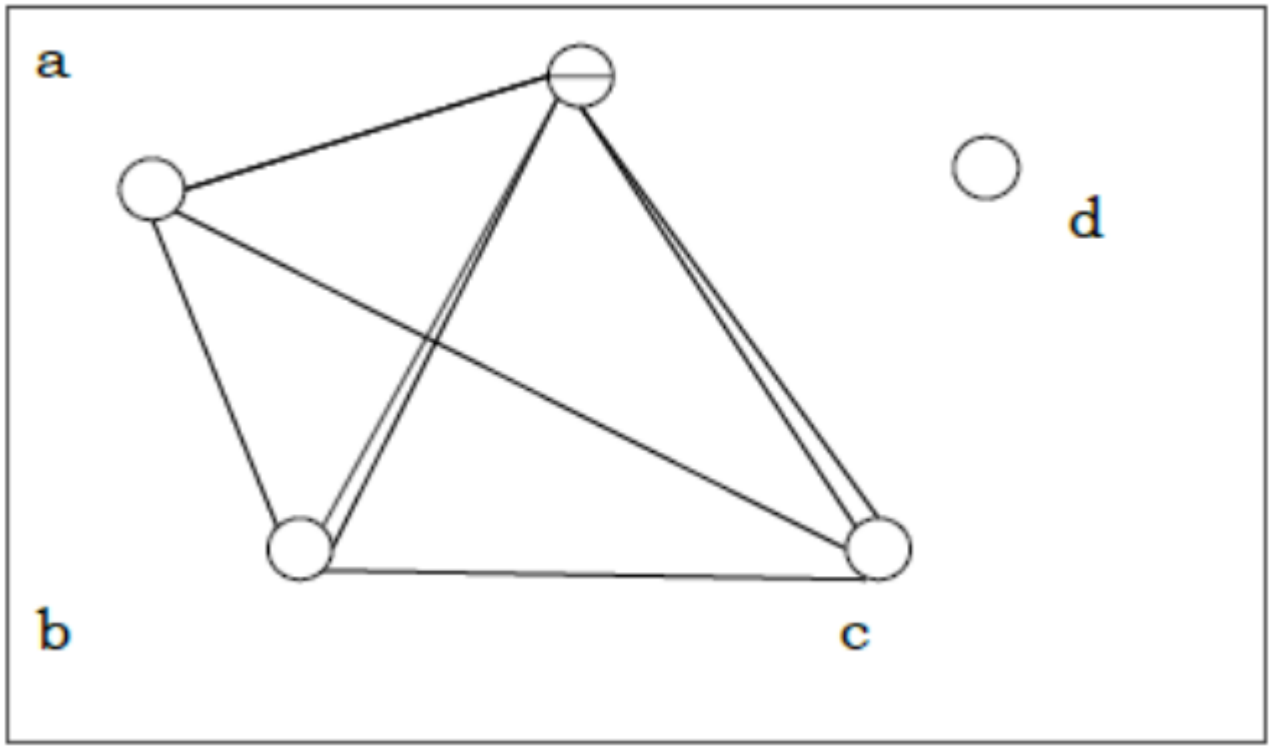
Planar graph – A graph G is called a planar graph if it can be drawn in a plane without any edges crossed. If we draw graph in the plane without edge crossing, it is called embedding the graph in the plane.



Planar Graph

Non-planar graph – A graph is non-planar if it cannot be drawn in a plane without graph edge crossing

Types of Graphs



Non Planar Graph

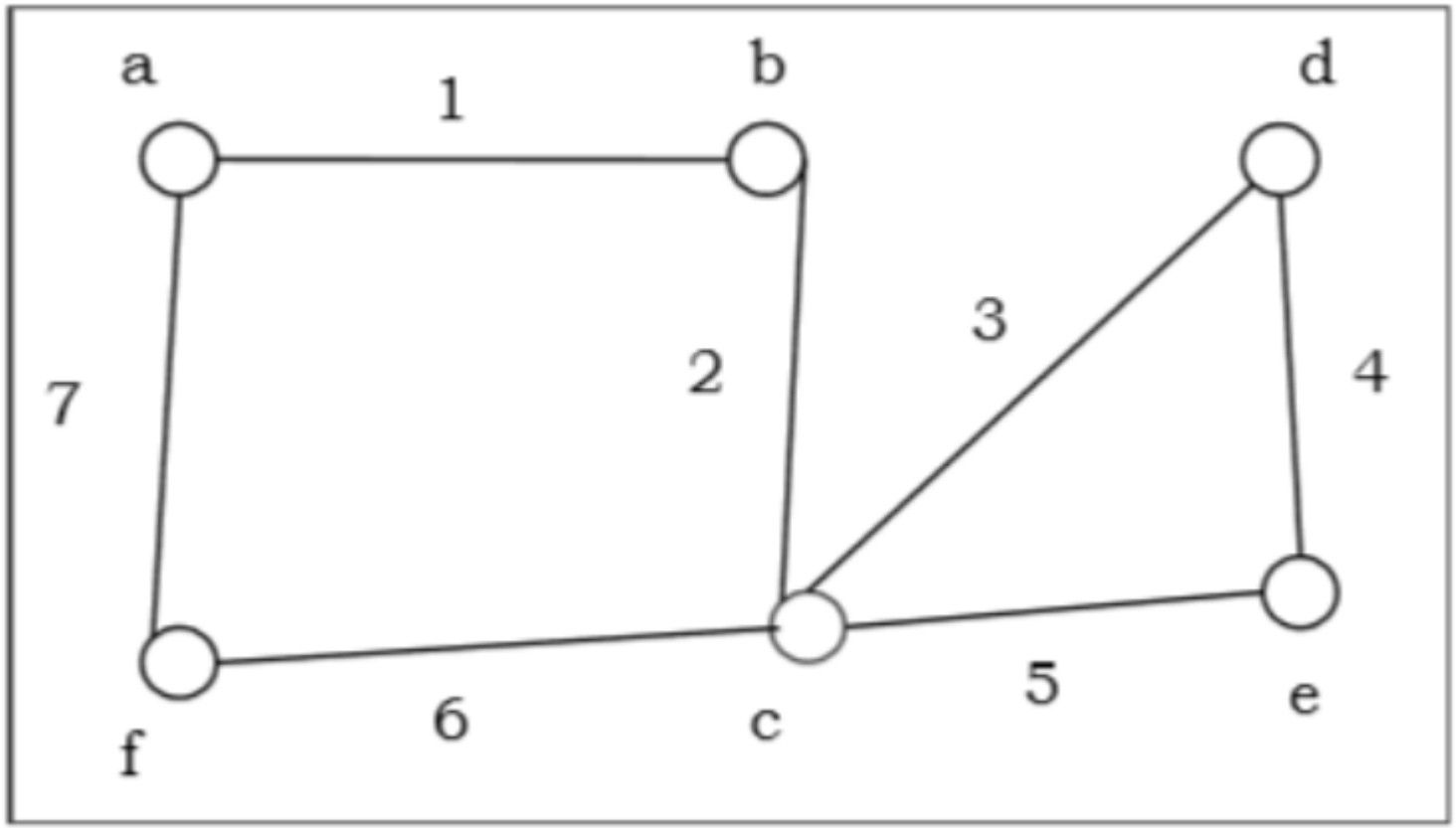
Types of Graphs

Euler Graphs

A connected graph G is called an Euler graph, if there is a closed trail which includes every edge of the graph G. An Euler path is a path that uses every edge of a graph exactly once. An Euler path starts and ends at different vertices.

An Euler circuit is a circuit that uses every edge of a graph exactly once. An Euler circuit always starts and ends at the same vertex. A connected graph G is an Euler graph if and only if all vertices of G are of even degree, and a connected graph G is Eulerian if and only if its edge set can be decomposed into cycles.

Euler Graphs



NECESSARY AND SUFFICIENT CONDITIONS FOR EULER CIRCUITS AND PATHS

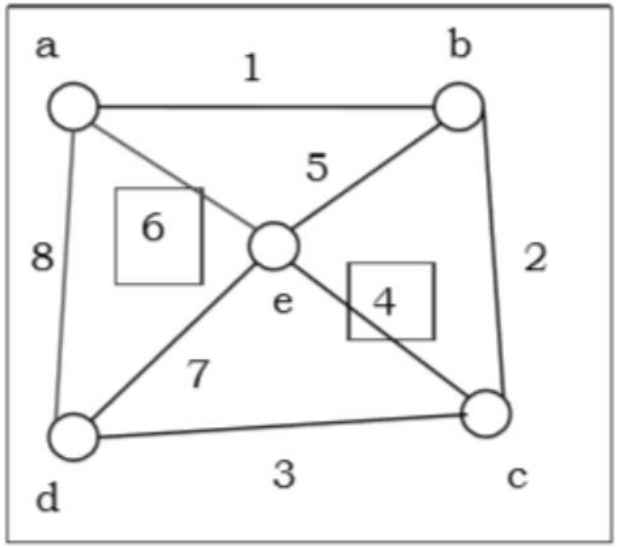
A connected multigraph with at least two vertices has an Euler circuit if and only if each of its vertices has even degree.

A connected multigraph has an Euler path but not an Euler circuit if and only if it has exactly two vertices of odd degree.

Hamiltonian Graphs

A connected graph G is called Hamiltonian graph if there is a cycle which includes every vertex of G and the cycle is called Hamiltonian cycle. Hamiltonian walk in graph G is a walk that passes through each vertex exactly once.

If G is a simple graph with n vertices, where $n \geq 3$ If $deg(v) \geq \frac{n}{2}$ for each vertex v, then the graph G is Hamiltonian graph. This is called **Dirac's Theorem**.



Hamiltonian Circuits conditions

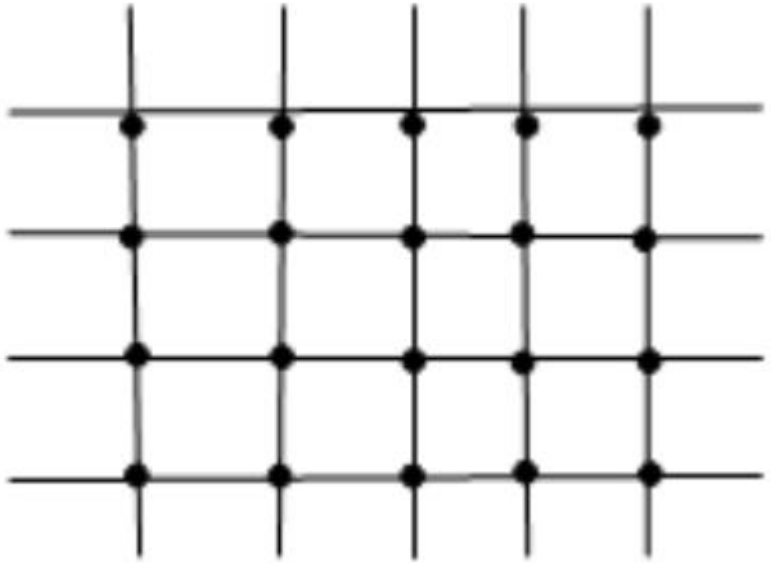
DIRAC'S THEOREM If G is a simple graph with n vertices with $n \geq 3$ such that the degree of every vertex in G is at least $n/2$, then G has a Hamilton circuit.

ORE'S THEOREM If G is a simple graph with n vertices with $n \geq 3$ such that $\deg(u) + \deg(v) \geq n$ for every pair of nonadjacent vertices u and v in G, then G has a Hamilton circuit.

Finite and Infinite Graph

If a set of vertices V in graph G is a finite set then it is called finite graph.

If a set of vertices of V in graph G is an infinite set then it is called infinite graph.



Representation of Infinite Graph

Graph Terminology

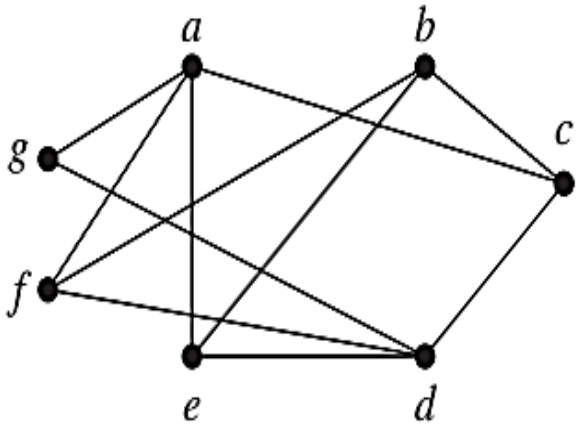
Two vertices u and v in an undirected graph G are called *adjacent* (or *neighbors*) in G if u and v are endpoints of an edge e of G. Such an edge e is called *incident with* the vertices u and v and e is said to *connect* u and v.

How many edges are there in a graph with 10 vertices each of degree six?

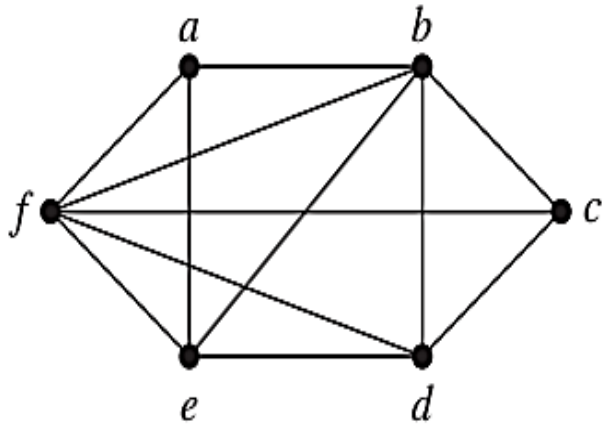
Solution: Because the sum of the degrees of the vertices is $6 \cdot 10 = 60$, it follows that $2m = 60$ where m is the number of edges. Therefore, $m = 30$.

Bipartite Graphs in detail

Are the graphs G and H displayed in Figure bipartite?



G



H

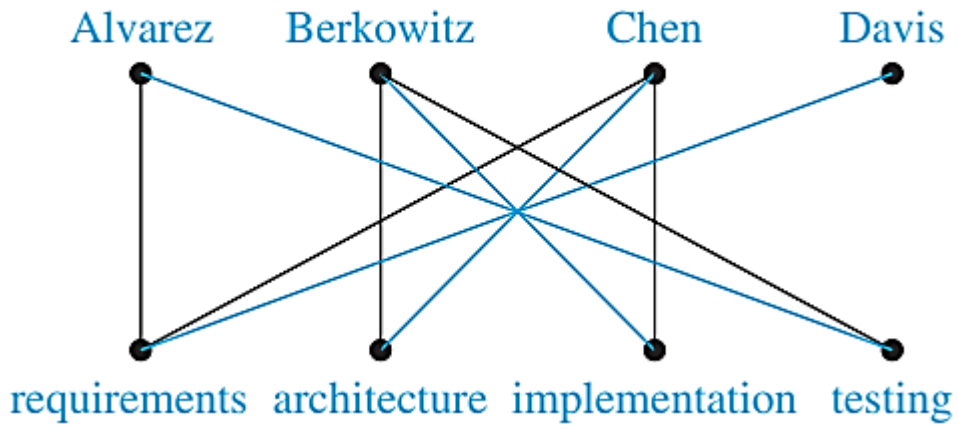
Solution: Graph G is bipartite because its vertex set is the union of two disjoint sets, {a, b, d} and {c, e, f, g}, and each edge connects a vertex in one of these subsets to a vertex in the other subset.
(Note that for G to be bipartite it is not necessary that every vertex in {a, b, d} be adjacent to every vertex in {c, e, f, g}. For instance, b and g are not adjacent.)

Graph H is not bipartite because its vertex set cannot be partitioned into two subsets so that edges do not connect two vertices from the same subset.

Bipartite Graphs and Matchings

Bipartite graphs can be used to model many types of applications that involve matching the elements of one set to elements of another,

First, suppose that a group has four employees: Alvarez, Berkowitz, Chen, and Davis; and suppose that four jobs need to be done to complete Project 1: requirements, architecture, implementation, and testing. Suppose that Alvarez has been trained to do requirements and testing; Berkowitz has been trained to do architecture, implementation, and testing; Chen has been trained to do requirements, architecture, and implementation; and Davis has only been trained to do requirements. We model these employee capabilities using the bipartite graph in Figure



To complete Project , we must assign an employee to each job so that every job has an employee assigned to it, and so that no employee is assigned more than one job.

We can do this by assigning Alvarez to testing, Berkowitz to implementation, Chen to architecture, and Davis to requirements, as shown in Figure

Representing Graphs

Adjacency Matrices

Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \ldots, v_n$. The **adjacency matrix** $\mathbf{A}$ (or $\mathbf{A}_G$) of G, with respect to this listing of the vertices, is the n x n zero–one matrix with 1 as its (i, j)th entry when v_i and v_j are adjacent, and 0 as its (i, j)th entry when they are not adjacent. In other words, if its adjacency matrix is $\mathbf{A} = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

Adjacency Matrices

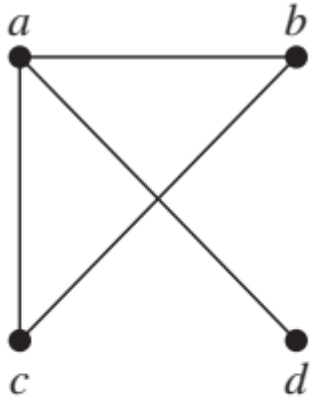


FIGURE 3
Simple Graph.

Use an adjacency matrix to represent the graph shown in Figure 3.

Solution: We order the vertices as a, b, c, d. The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Adjacency Matrices

Draw a graph with the adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d.

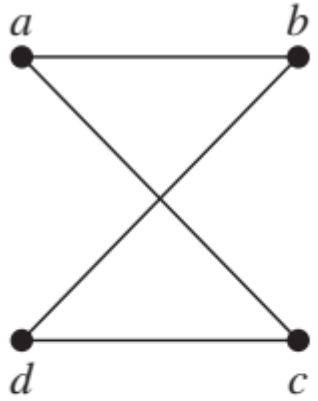


FIGURE 4
A Graph with the Given Adjacency Matrix.

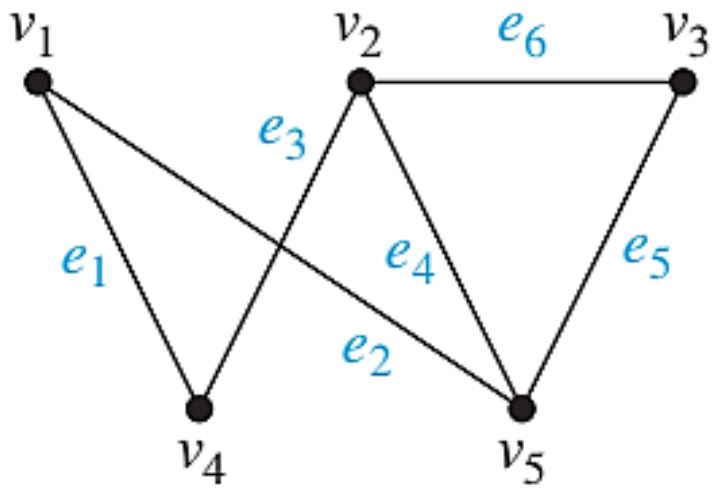
Incidence Matrices

Another common way to represent graphs is to use **incidence matrices**. Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \ldots, v_n$ are the vertices and $e_1, e_2, \ldots, e_m$ are the edges of G. Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix $\mathbf{M} = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

Incidence Matrices

Represent the graph shown in Figure with an incidence matrix.

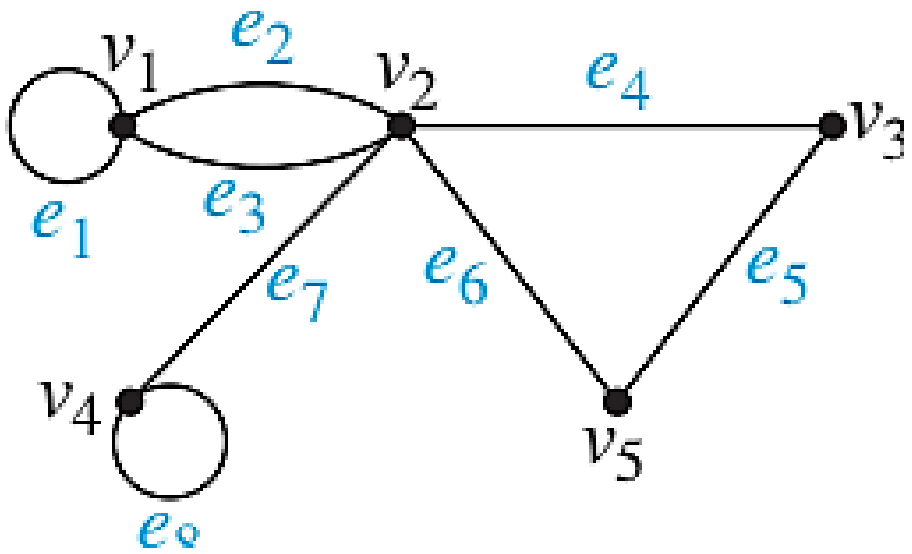


Solution: The incidence matrix is

	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	1	0	0	0	0
v_2	0	0	1	1	0	1
v_3	0	0	0	0	1	1
v_4	1	0	1	0	0	0
v_5	0	1	0	1	1	0

Incidence Matrices

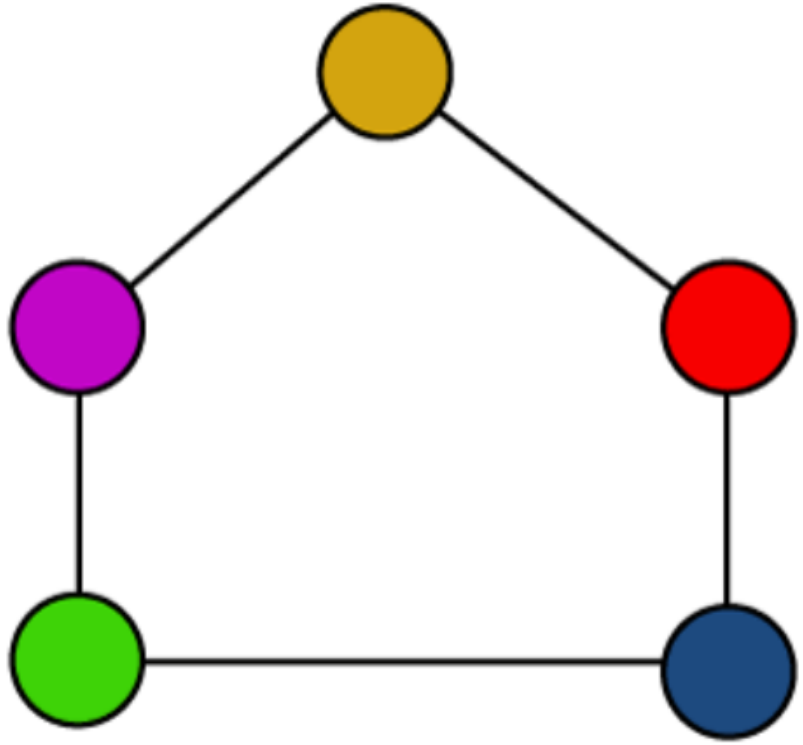
Represent the graph shown in Figure with an incidence matrix.



$$
\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array}
\begin{array}{c}
\begin{array}{cccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{array} \\
\left[\begin{array}{cccccccc}
1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 1 & 1 & 0 & 0
\end{array}\right]
\end{array}.
$$

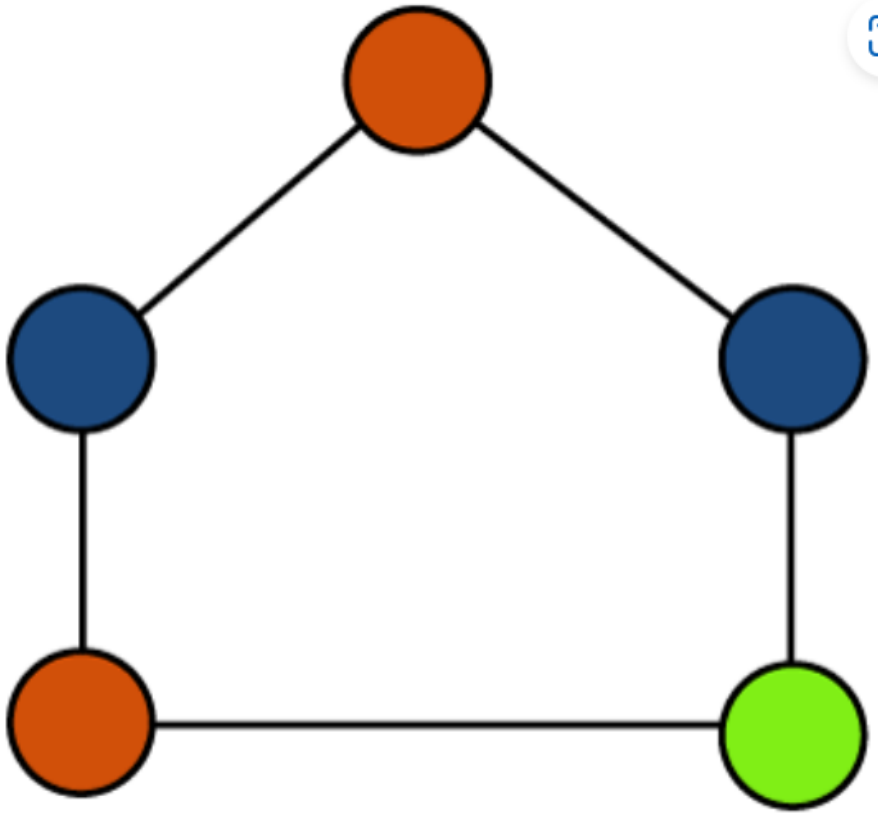
Graph Coloring

Graph coloring can be described as a process of assigning colors to the vertices of a graph. In this, the same color should not be used to fill the two adjacent vertices. We can also call graph coloring as Vertex Coloring. In graph coloring, we have to take care that a graph must not contain any edge whose end vertices are colored by the same color. This type of graph is known as the Properly colored graph.



Chromatic Number

The chromatic number can be described as the minimum number of colors required to properly color any graph. In other words, the chromatic number can be described as a minimum number of colors that are needed to color any graph in such a way that no two adjacent vertices of a graph will be assigned the same color.



The above graph contains some points, which are described as follows:

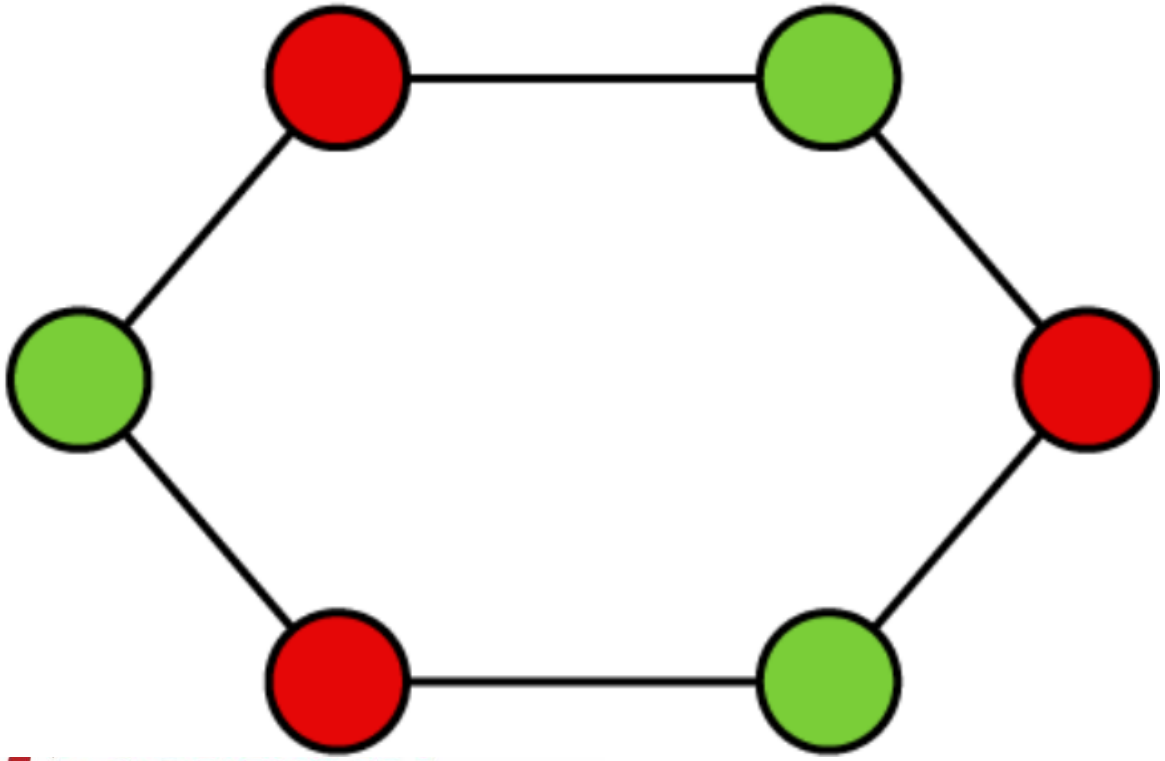
The same color is not used to color the two adjacent vertices.
The minimum number of colors of this graph is 3, which is needed to properly color the vertices.

Hence, in this graph, the chromatic number = 3
If we want to properly color this graph, in this case, we are required at least 3 colors.

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Chromatic Number

In the following graph, we have to determine the chromatic number.



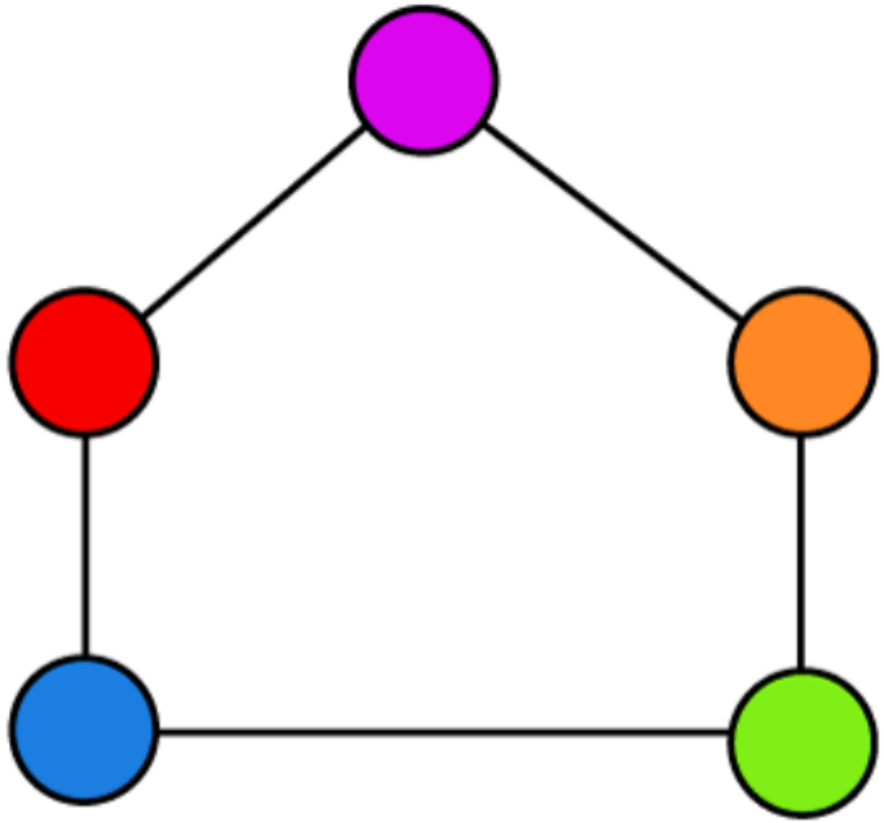
In the above graph, there are 2 different colors for six vertices, and none of the edges of this graph cross each other. So

Chromatic number = 2

Here, the chromatic number is less than 4, so this graph is a plane graph.

Chromatic Number

In the following graph, we have to determine the chromatic number.



In the above graph, there are 2 different colors for six vertices, and none of the edges of this graph cross each other. So

Chromatic number = 5

Here, the chromatic number is more than 4, so this graph is a non plane graph.

Isomorphism of Graphs

We often need to know whether it is possible to draw two graphs in the same way. That is, do the graphs have the same structure when we ignore the identities of their vertices?

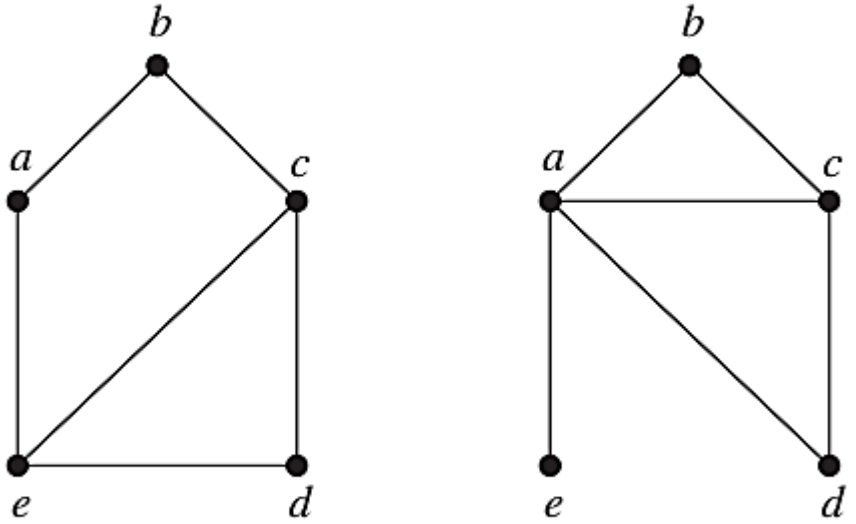
The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are *isomorphic* if there exists a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2, for all a and b in V_1. Such a function f is called an *isomorphism*.* Two simple graphs that are not isomorphic are called *nonisomorphic*.

Isomorphism of Graphs

- A property preserved by isomorphism of graphs is called a graph invariant. For instance, isomorphic simple graphs must have the same number of vertices, because there is a one-to-one correspondence between the sets of vertices of the graphs.
- Isomorphic simple graphs also must have the same number of edges, because the one-to-one correspondence between vertices establishes a one-to-one correspondence between edges.
- In addition, the degrees of the vertices in isomorphic simple graphs must be the same.
- That is, a vertex v of degree d in G must correspond to a vertex f (v) of degree d in H, because a vertex w in G is adjacent to v if and only if f (v) and f (w) are adjacent in H.

Isomorphism of Graphs

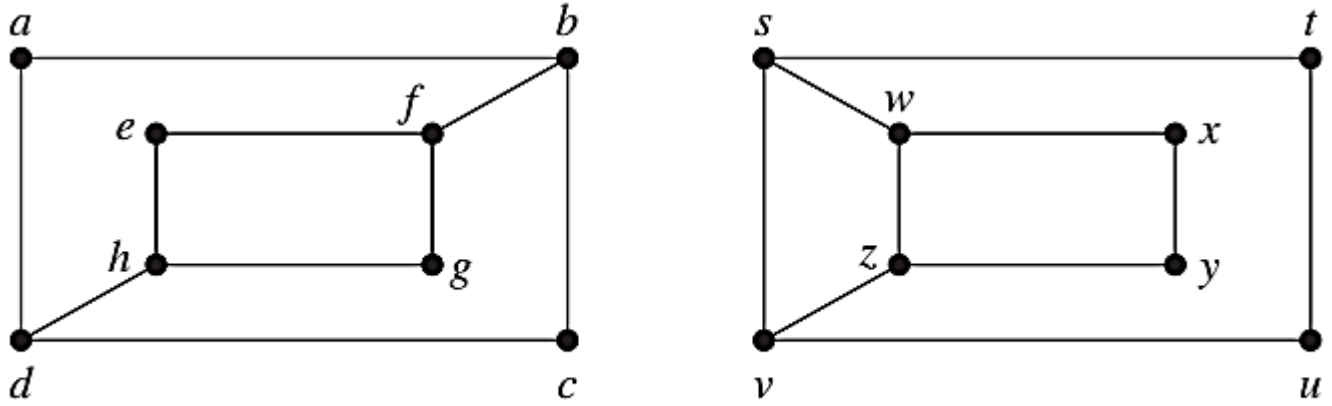
Show that the graphs displayed in Figure are not isomorphic.



Both G and H have five vertices and six edges. However, H has a vertex of degree one, namely, e, whereas G has no vertices of degree one. It follows that G and H are not isomorphic.

Isomorphism of Graphs

Determine whether the graphs shown in Figure are isomorphic.



- The graphs G and H both have eight vertices and 10 edges. They also both have four vertices of degree two and four of degree three. Because these invariants all agree, it is still conceivable that these graphs are isomorphic.
- However, G and H are not isomorphic. To see this, note that because deg(a) = 2 in G, a must correspond to either t, u, x, or y in H, because these are the vertices of degree two in H. However, each of these four vertices in H is adjacent to another vertex of degree two in H, which is not true for a in G.

THANK YOU